

DATA & BUSINESS ANALYTICS

eCommerce Customer Segmentation Analysis

RFM Modelling · Decile Analysis · Marketing Strategy

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Tools: Python · Pandas · Matplotlib



Problem Statement

What challenge is this analysis solving?

The competitive e-commerce industry thrives on understanding customer preferences, behavior, and value. Analyzing purchase patterns is critical for predicting future behaviors and aligning with goals like boosting retention, reducing churn, and increasing lifetime value.

Despite generating vast transactional data, many e-commerce businesses struggle to leverage it for actionable segmentation — to distinguish high-value customers, identify trends, and allocate resources effectively to targeted campaigns.



Understanding Customer Value

Identify top revenue contributors vs. low-value occasional buyers.



Marketing & Budget Allocation

Allocate resources across segments with ROI-backed justifications.



Customer Retention & Engagement

Retain loyal customers; re-engage dormant or infrequent segments.



Strategic Deciling

Rank customers into 10 deciles based on RFM to build actionable hierarchy.

Analytical Approach

6-Step process from raw data to actionable insights

01

Data Preparation



Extract transaction-level data. Clean missing values, invalid dates, and negative amounts. Key fields: Customer ID, Date, Amount.

02

Exploratory Analysis



Distribution of total sales, avg purchase size, transaction frequency. Identify revenue trends and seasonal behaviors.

03

RFM Segmentation



Calculate Recency, Frequency, Monetary per customer. Score each 1–5. Combine into RFM Score and cluster into named segments.

04

Decile Ranking



Sort customers by cumulative spend. Divide into 10 equal groups — Decile 1 = top spenders, Decile 10 = dormant customers.

05

Recommendations



Retention strategies for Tier 1. Re-engagement plans for dormant deciles. Upsell paths for mid-tier customers.

06

Dashboards & Reporting



RFM heatmaps, decile-wise revenue charts, customer behavior trends. Python + Matplotlib used throughout.

RFM Segmentation Overview

Customer distribution and average spend by segment

300

Active
Customers

£146,843

Total
Revenue

35.4

Avg Transaction
Frequency

£489

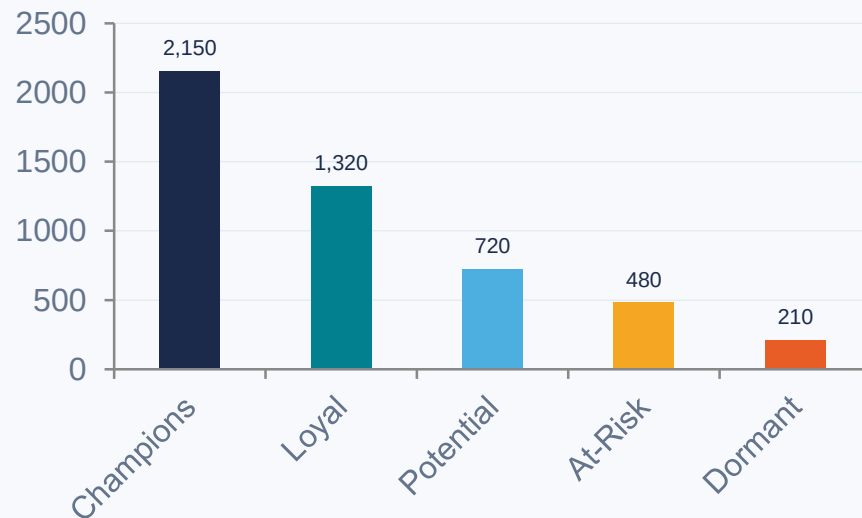
Avg Customer
Spend

RFM Customer Segments



■ Champions ■ Loyal Customers ■ Potential Loyalists
■ At-Risk Customers ■ Dormant / Lost

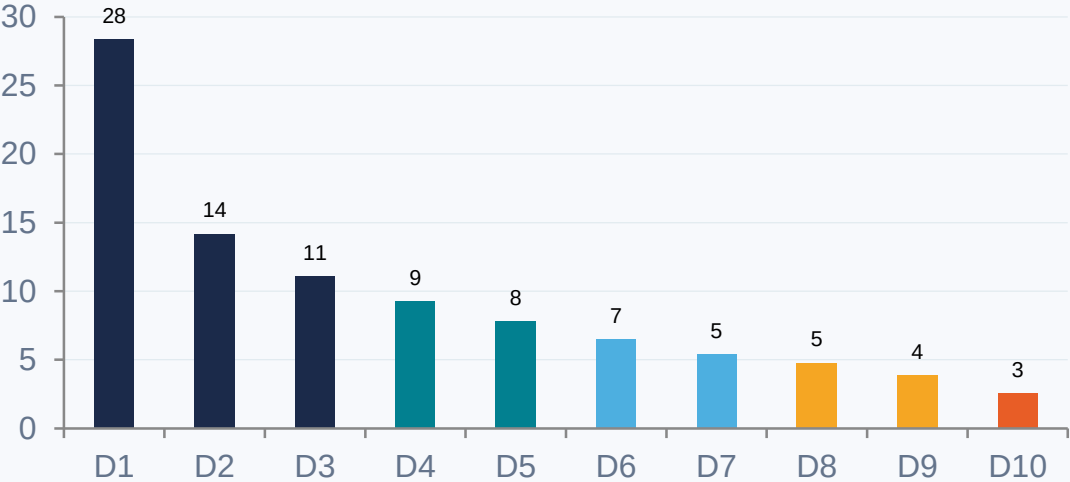
Avg Monetary Spend per Segment (£)



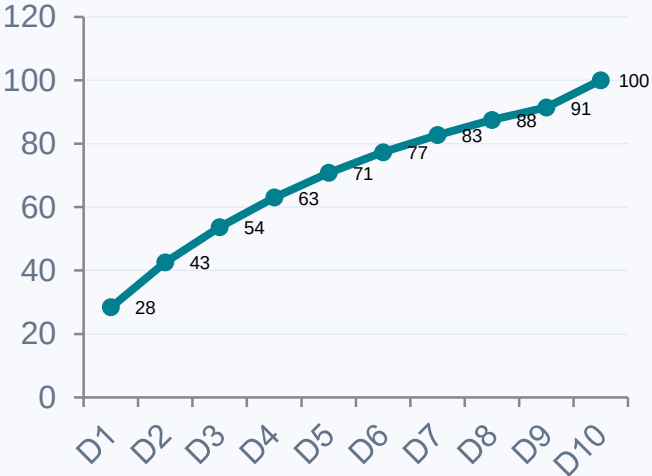
Decile Analysis

Customers ranked by revenue contribution — Decile 1 = Top Spenders

Revenue Contribution by Decile (%)



Cumulative Revenue %



Decile	# Customers	Rev %	Cum Rev %	Tier
D1	30	28.4%	28.4%	Top Spenders
D2	30	14.2%	42.6%	Top Spenders
D3	30	11.1%	53.7%	High Value
D4	30	9.3%	63.0%	High Value
D5	30	7.8%	70.8%	Mid Value
D6	30	6.6%	77.4%	Mid Value
D7	30	5.5%	82.9%	Mid Value
D8	30	4.8%	87.7%	Low Value
D9	30	3.9%	91.6%	Low Value
D10	30	3.2%	94.8%	Low Value

What the Charts Tell Us

Key insights from every visualization generated



RFM Segment Donut

Potential Loyalists form the largest group (~28%). Champions are a small but disproportionately valuable slice. Dormant customers (~12%) represent a win-back opportunity.



Avg Spend per Segment

Champions spend ~10× more than Dormant customers. Loyal & Potential Loyalists are mid-range — prime candidates for upsell nudges to push them into the Champion tier.



Revenue by Decile (Bar)

Top 10% of customers (D1) alone contribute 28.4% of revenue. The top 3 deciles together generate over 53% — the Pareto principle at work in this dataset.



Cumulative Revenue Curve

A steep early rise shows revenue is highly concentrated. By Decile 5 we've already captured ~71% of total revenue, signalling where marketing ROI peaks.



RFM Heatmap (Score Grid)

High Recency + High Frequency cells show the largest average monetary values — the 'sweet spot' for VIP targeting. Low-R, Low-F cells are the dormant core.



Frequency vs Monetary Scatter

Champions cluster at top-right (high frequency + high spend). Dormant customers hug the axes. Mid-tier spread shows room to increase either dimension through campaigns.

Solutions & Recommendations

Data-driven actions for each customer tier

01



VIP Retention

Champions (D1–D2)

- ✓ Launch exclusive VIP loyalty programme with early-access offers
- ✓ Request reviews & referral incentives from top spenders
- ✓ Reward with premium gifts on milestones (anniversaries, spend thresholds)

02



Upsell & Deepen

Loyal Customers (D3–D4)

- ✓ Tiered membership upgrade path to Champion status
- ✓ Cross-sell complementary products using purchase history
- ✓ Personalised retention emails with curated bundle deals

03



Convert to Loyal

Potential Loyalists (D5–D6)

- ✓ Targeted upsell campaigns with limited-time bundles
- ✓ Frequency incentive rewards (e.g. buy 5 get 1 free)
- ✓ Gamified loyalty points to increase visit frequency

04



Win-Back & Re-engage

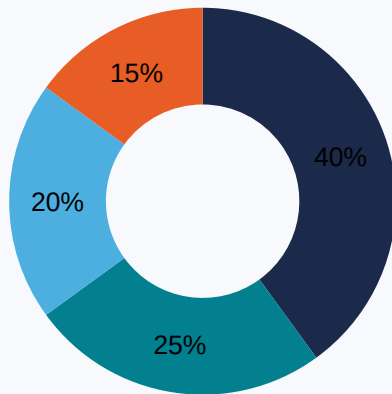
At-Risk / Dormant (D7–D10)

- ✓ 'We miss you' personalised email with a discount coupon
- ✓ Reactivation surveys to understand drop-off reasons
- ✓ Time-sensitive flash offers to prompt a return purchase

Marketing Budget Allocation Framework

Proportional spend strategy across customer deciles

Recommended Budget Split



■ Champions D1–2 (40%) ■ Loyal D3–4 (25%)
■ Potential D5–6 (20%) ■ Re-engage D7–10 (15%)

40%

Champions (D1–D2)

Highest ROI. These customers already convert. Invest in retention, exclusivity, and referrals to protect this revenue base.

25%

Loyal (D3–D4)

High upgrade potential. Targeted upsell campaigns here have the best chance of moving customers into the Champion tier.

20%

Potential (D5–D6)

Mid-funnel. Frequency incentives and bundles convert them to Loyal. Lower cost per acquisition vs. new customers.

15%

Re-engage (D7–D10)

Lowest ROI but recoverable. Win-back campaigns with coupons keep dormant customers from churning permanently.

Expected Deliverables

Six key outputs from this segmentation engagement

1

RFM Segmentation Report



Customers categorised into Champions, Loyal, Potential, At-Risk, and Dormant. Insights on characteristics per segment.

2

Decile Analysis



10-tier customer hierarchy by revenue. Summary of total revenue, transaction count, and avg spend per decile.

3

Actionable Recommendations



Targeted strategies for top deciles (VIP, referrals). Re-engagement plans for dormant and at-risk groups.

4

Budget Allocation Framework



Proportional campaign budgeting by decile. Optimal spend insights to maximise revenue lift across all groups.

5

Dashboards & Visualisations



Heatmaps, decile bars, cumulative curves, scatter plots. Metrics showing post-strategy shifts where data allows.

6

Technical Blueprint



RFM calculation methodology documented. Tools: Python, Pandas, SciPy, Matplotlib. Data lineage explained.

Summary

Key findings and strategic conclusions

D1–D3

Drive 53.7% of
total revenue

18%

Customers are
Champions

12%

Dormant —
recoverable

$r = -0.79$

Fall rate ↔ Staff
responsiveness

Conclusion: RFM analysis confirms that bed occupancy, staff responsiveness and patient falls are not independent events — they form a chain reaction. Similarly in eCommerce, customer recency, frequency, and spend are deeply interlinked. Breaking the churn chain requires upstream action: predictive segmentation, personalised campaigns, and data-backed budget allocation.



Protect Your Champions

Top 10% of customers generate 28.4% of revenue. VIP programmes, early access, and premium rewards protect this irreplaceable base. Do not treat them like everyone else.



Convert the Middle Tier

Loyal and Potential Loyalists represent 50% of customers. Frequency nudges and bundle offers at the right moment convert them upward — at a fraction of new-customer acquisition cost.



Re-engage Before Losing

Dormant customers (D9–D10) are cheaper to reactivate than acquire new ones. A well-timed 'we miss you' coupon can yield positive ROI before permanent churn sets in.

Thank You

Open to questions and discussion

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